

explicitly compute the derivatives of (10) as long as these derivatives converge. For arbitrary m we have

$$\begin{aligned}
\frac{\partial I_n}{\partial \delta_m} &= \frac{\partial}{\partial \delta_m} \sum_{i,j=0}^{n+1} \sum_{k=1}^{\infty} 2^{q_i+q_j} (-1)^{i+j} \frac{\Delta_{ij}^k}{k!k} z_c^k \\
&= \frac{\partial}{\partial \delta_m} \left\{ \sum_{i=0}^{n+1} \sum_{k=1}^{\infty} 2^{q_i+q_m} (-1)^{i+m} \frac{\Delta_{im}^k}{k!k} z_c^k \right. \\
&\quad \left. + \sum_{i=0}^{n+1} \sum_{k=1}^{\infty} 2^{q_m+q_i} (-1)^{m+i} \frac{\Delta_{mi}^k}{k!k} z_c^k \right\} \\
&= \sum_{i=0}^{n+1} \sum_{k=1}^{\infty} 2^{q_m+q_i} (-1)^{m+i} \frac{z_c^k}{k!} i [\Delta_{mi}^{k-1} - (-1)^{k-1} \Delta_{mi}^{k-1}].
\end{aligned} \tag{11}$$

The terms with k odd cancel, so the result can be simplified as

$$\begin{aligned}
&= \sum_{i=0}^{n+1} \sum_{k=1}^{\infty} 2^{q_m+q_i+1} (-1)^{m+i} \frac{z_c^{2k}}{(2k)!} i^{2k} (\delta_m - \delta_i)^{2k-1} \\
&= \sum_{i \neq m}^{n+1} \frac{1}{\delta_m - \delta_i} 2^{q_m+q_i+1} (-1)^{m+i} \sum_{k=1}^{\infty} \frac{z_c^{2k}}{(2k)!} i^{2k} (\delta_m - \delta_i)^{2k} \\
&= \sum_{i \neq m}^{n+1} \frac{1}{\delta_m - \delta_i} 2^{q_m+q_i+1} (-1)^{m+i} \{ \cos[(\delta_m - \delta_i) z_c] - 1 \}.
\end{aligned} \tag{12}$$

Here we have used the expansion $\cos(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}$

which converges on the whole complex plane. From (12) we know that the derivatives of I_n indeed converge to a finite value. Thus the optimal pulse locations $\{\delta_1, \delta_2, \dots, \delta_n\}$ shall satisfy the following non-linear equations

$$\sum_{i \neq m}^{n+1} \frac{1}{\delta_m - \delta_i} 2^{q_m+q_i+1} (-1)^{m+i} \{ \cos[(\delta_m - \delta_i) z_c] - 1 \} = 0. \tag{13}$$

Equations (13) are main results of this paper. The optimal sequence obtained from (13) is quite different from the UDD sequence obeying

$$\sum_{j=1}^{n+1} 2^{q_j} (-1)^j \delta_j^p = 0$$

for $p = \{1, 2, \dots, n\}$. For the ohmic spectrum, our equations incorporate the UV cutoff frequency ω_c , indicating that the solutions are specially tailored to combat this kind of noise. Although the UDD sequence is universal in suppressing decoherence, we believe that the HLODD sequence will outperform the UDD sequence in the ohmic environment. In the next section, we use numeric methods to illustrate the performance of HLODD sequence.

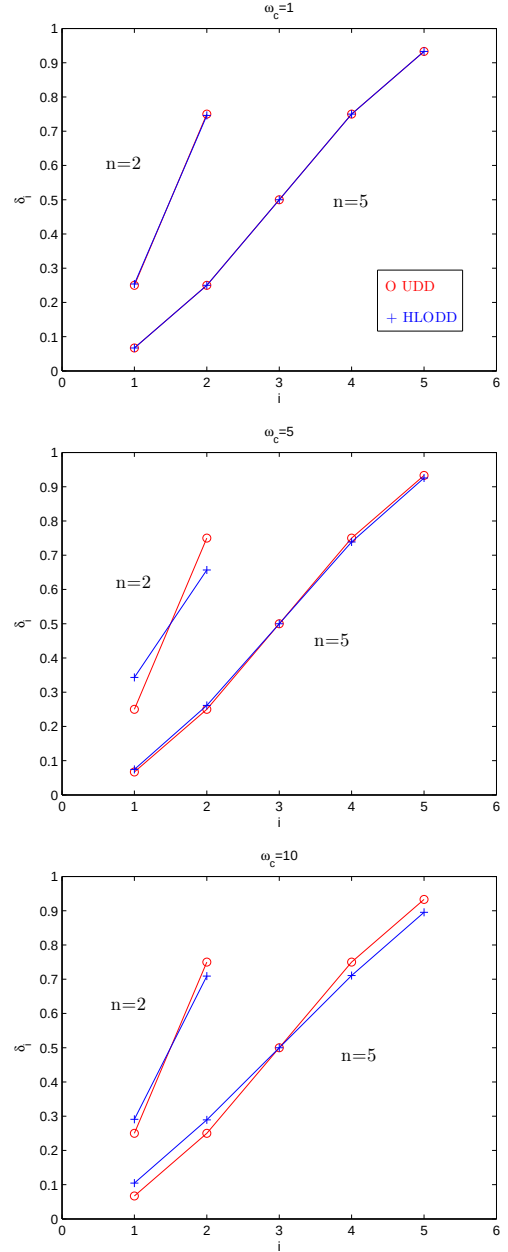


FIG. 1: Comparison between UDD and HLODD sequence for different UV cutoff frequency ω_c . Pulse sequences δ_i for $n = 2$ and $n = 5$ are plotted in one figure under the same ω_c .

IV. NUMERICAL RESULTS

We start our simulation by solving the non-linear equations (13) and evaluate the decoherence function with these solutions. First, we set the total evolution time $t = 1$ without loss of generality. Then $z_c = \omega_c$ and we can concentrate on analyzing the influence of the cutoff frequency ω_c . Computing solutions to (13) for different ω_c , we find that the optimal pulse sequences behave differently. We also evaluate the UDD sequence for comparison.